

What Are You Proving?

A proof statement — FlyRank AI Backend and AI Engineer Internship, Week 1

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The Proof Statement

ONE CLAIM

I integrate LLMs into working backend systems: RAG pipelines, semantic search, and AI-powered features built on real production stacks like Django, FastAPI, and PostgreSQL.

ONE PERSON

A technical hiring manager reviewing my GitHub before deciding whether to schedule a call.

ONE ACTION

Look at the repos.

Full Statement

I integrate LLMs into working backend systems, specifically RAG pipelines, semantic search, and AI-powered features built on real production stacks like Django, FastAPI, and PostgreSQL. This is written for a technical hiring manager reviewing my GitHub before deciding whether to schedule a call. The one action I want them to take is simple: look at the repos. DocuMind shows a full RAG pipeline (ChromaDB, Sentence Transformers, Groq LLM) with citation-grounded answers. Job Connect shows the same pattern applied to semantic resume matching. BE-04 in my FlyRank internship repo shows a repository pattern that swaps storage backends without touching route logic — the same discipline that makes swapping model providers just as clean.

The One Honest Line

A CV can list “RAG” and “LLMs” as keywords, but it can’t prove I actually built one that works, ran into real failure modes, and fixed them. Only a working repo with commits and screenshots can do that.

Why This Needs to Exist

A resume can claim skills. A portfolio with no clear claim is just decoration — a list of projects with no argument for why any one of them matters to the person reading it. The three supporting projects in this document were chosen deliberately, not because they are the newest or the flashiest, but because together they prove one narrow, specific thing: that RAG pipelines and LLM integration are not something I read about, but something I built, broke, and fixed inside real backend systems.

Each of the following sections maps directly back to a piece of the claim above.

Claim component	Supporting evidence
RAG pipeline, grounded in real documents	DiscuMind — PDF upload, chunking, embeddings, ChromaDB, Groq LLM, citations
Semantic search applied to a real product	Job Connect AI — resume parsing, ATS scoring, semantic job matching
Architecture discipline that generalizes	BE-04 (Flynn and others) — repository pattern, Postgres in Docker, persistence p

Evidence 1 — DocuMind

AI Legal & Financial Document Analyst

Stack: Django, ChromaDB, Sentence Transformers, Groq LLM, RAG, pypdf, PostgreSQL

This is the project that most directly proves the RAG half of the claim. It was built specifically to close a gap flagged in two real interviews: hands-on experience with retrieval-augmented generation, vector databases, and LLM orchestration, not just theoretical familiarity with the terms.

- Full RAG pipeline: PDF upload, text extraction, intelligent chunking, Sentence Transformer embedding generation, ChromaDB vector storage, and semantic retrieval scoped per user document.
- Groq LLM integrated with a strict grounded prompt so answers return with source citations and chunk previews — the model is not allowed to answer outside the uploaded document's context.
- Persistent document chat history with user-scoped isolation, SQLite for local development, and PostgreSQL readiness for production via environment-driven configuration.

Evidence 2 — Job Connect

AI-Powered Full-Stack Job Portal (Final Year Project)

Stack: Django, DRF, Django Channels, WebSockets, Redis, ChromaDB, Sentence Transformers, Groq LLM, PostgreSQL, Render

This project proves the claim generalizes past a single-purpose RAG tool into a full production application with real users, real-time features, and AI woven into the core product logic rather than bolted on as a demo.

- Full-stack AI-powered job portal supporting Job Seekers, Employers, and Admins, with role-based authentication, resume management, job posting, applications, and real-time chat via Django Channels and WebSockets.
- AI Resume Intelligence: resume parsing, ATS scoring, career analysis, semantic job matching using Sentence Transformers and ChromaDB, and AI-generated personalized cover letters powered by Groq LLM.
- A recommendation engine combining vector similarity search with structured scoring to recommend relevant jobs, deployed with Redis, PostgreSQL, and ChromaDB persistence on Render.

Evidence 3 — BE-04 (FlyRank Internship)

Containerize Your Stack

Stack: FastAPI, PostgreSQL, Docker, Docker Compose, Repository Pattern

This one is not an AI project on its surface, and that is exactly why it belongs here. It proves the architecture discipline underneath the claim: the same repository-pattern thinking that lets a storage backend be swapped without touching route logic is what makes swapping an LLM provider, or a vector store, an equally clean change rather than a rewrite.

- Postgres running in Docker with a named volume, connected via a PostgresRepository that implements the exact same interface as the original in-memory store.
- main.py picks the backend based on a single environment variable — routes and service logic never branch on which storage implementation is active.
- Persistence proven three separate ways: after an app-only restart, after a full docker compose down/up, and via a browser test on the forwarded port.

Repository: github.com/samlakhan7/Flyrank-Backend-Tasks/tree/main/BE-04

View the Repos

GitHub: github.com/salamlakhan7

FlyRank internship repo: github.com/salamlakhan7/Flyrank-Backend-Tasks

LinkedIn: [linkedin.com/in/abdul-salam-501b2025b](https://www.linkedin.com/in/abdul-salam-501b2025b)

This document exists because a resume can claim “RAG” and “LLMs,” but only a working repository with real commits, real screenshots, and real failure modes fixed along the way can prove it.